

TORSION OF THE PERFECT-CUBOID ELLIPTIC FAMILY IS DEGENERATE FOR THE CUBOID RECOVERY MAP

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ABSTRACT. For a Pythagorean rational q (one with $1 + q^2$ a rational square) the perfect-cuboid attack attaches to the corresponding fibre the elliptic curve $E_{\text{PCP}}(q): Y^2 = X(X + 1)(X + q^2)$ together with a rational recovery map $\varphi(X, Y) = 2Yq/(q^2 - X^2)$ that returns the candidate cuboid edge of a rational point. The rational torsion of this family is $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. This classification is not new: in substance and method (Mazur's theorem followed by a Fermat descent on $u^2 = s^4 + 1$) it is that of Yoshida's Lemma 2.1 for his face-cuboid family $E_{1,s}$, which we credit accordingly and re-derive through an explicit uniform discriminant identity $\Delta(Z) = (Z - 1)^4(Z + 1)^2(Z^2 - 6Z + 1)$; we also record that $E_{\text{PCP}}(q)$ and $E_{1,s}$ are not $\mathbb{Q}$ -isomorphic on the Pythagorean locus, only twist-related. The contribution of this note is one statement, with no analogue in Yoshida's work, on the interaction of the torsion with the recovery map: every non-identity rational torsion point of $E_{\text{PCP}}(q)$ is sent by φ into $\{0, \infty\}$, the three two-torsion points to 0 and the four order-four points to a pole (the identity being the point at infinity). Hence a torsion point never decodes to a finite nonzero edge, and any perfect-cuboid point on the fibre is of infinite order. The degeneracy is proved as an identity in $\mathbb{Q}(q)$ and confirmed numerically across 62 Pythagorean values. We offer it as a companion structural remark to Yoshida and as the lemma on which a Chabauty- or descent-type search of the non-torsion locus would rest. No claim is made about the existence of perfect cuboids.

1. INTRODUCTION

A *perfect cuboid* is a rectangular box whose three edges, three face diagonals, and space diagonal are all rational; whether one exists is a long-standing open question [1]. A productive way to organise the search is to slice the perfect-cuboid locus into one-parameter fibres indexed by a Pythagorean rational q (a rational with $1 + q^2$ a rational square) and to attach to each fibre an elliptic curve whose rational points carry the arithmetic of that fibre. In the parametrisation used here the relevant curve is

$$E_{\text{PCP}}(q): Y^2 = X(X + 1)(X + q^2), \quad q \in \mathbb{Q}^* \setminus \{0, \pm 1\}, \quad (1)$$

and a rational point (X, Y) is decoded back to a candidate cuboid edge by the *recovery map*

$$\varphi: E_{\text{PCP}}(q) \dashrightarrow \mathbb{P}^1, \quad \varphi(X, Y) = \frac{2Yq}{q^2 - X^2}. \quad (2)$$

A non-degenerate cuboid requires this value to be finite and nonzero.

The torsion of the family (1) is $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, uniformly in q . This classification is not new. Yoshida [3], studying *face* cuboids, introduces a family $E_{1,s}: y^2 = x(x - (2s)^2)(x + (s^2 - 1)^2)$ (a sub-family of Yagi's curves $y^2 = x(x - a^2)(x + b^2)$ with $a^2 + b^2$ a square) and proves in his Lemma 2.1 that $E_{1,s}(\mathbb{Q})_{\text{tor}} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ for all $s \in \mathbb{Q} \setminus \{0, \pm 1\}$, by Mazur's theorem followed by a descent that lands on $y^2 = x^3 - x$. Our family $E_{\text{PCP}}(q)$ obeys the same conclusion by the same route; the path differs only in bookkeeping, the descent here terminating on the rank-zero curve $y^2 = x^3 + 4x$ (Cremona 32a1) via the equation $u^2 = s^4 + 1$, that is, Fermat's theorem on right triangles. We

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therefore state the torsion classification as a recalled proposition and attribute it to Yoshida; the only presentational addition is an explicit uniform discriminant identity that isolates the order-eight obstruction directly in the parameter q .

It is worth recording, since an earlier draft of the present framework asserted otherwise, that $E_{\text{PCP}}(q)$ and $E_{1,s}$ are *not* $\mathbb{Q}$ -isomorphic on the Pythagorean locus. Their j -invariants coincide only at the non-Pythagorean value $q = (s^2 + 1)/(2s)$; for genuinely Pythagorean $q = (t^2 - 1)/(2t)$ no rational s matches the j -invariant of $E_{\text{PCP}}(q)$, as a search over s of bounded height confirms. The two families are related by a -1 quadratic twist rather than by isomorphism. The torsion conclusion survives this distinction because it rests on the rational two-torsion (three rational roots) and a rational point of order four, features the twist shares; thus Yoshida's Lemma 2.1 is the correct prior art for the *classification and its method*, applied here to a twist-inequivalent curve.

The genuinely new content of this note is one statement about the interaction of the torsion with the recovery map (2), a direction absent from Yoshida's paper, whose theorems run the opposite way (he uses non-torsion points to *construct* infinitely many face cuboids). We show that the recovery map is blind to torsion: every torsion point is degenerate. The three two-torsion points map to the edge value 0, and the four order-four points, sitting exactly on the pole divisor $X = \pm q$ of φ , map to a pole. Hence no torsion point yields a finite nonzero edge, and any perfect-cuboid point on the fibre must be of infinite order. The result is elementary once the torsion structure is in hand, and we make no claim beyond it: it does not resolve the perfect-cuboid problem, and it is offered as a structural companion to [3] and as the foundational lemma for a subsequent rank- or Chabauty-type analysis of the non-torsion locus.

2. THE CURVE AND ITS TORSION

We collect the standard facts, all uniform in q .

2.1. Rational two-torsion. Expanding (1) gives $Y^2 = X^3 + (1 + q^2)X^2 + q^2X$. The right side factors over $\mathbb{Q}$ as $X(X + 1)(X + q^2)$, so the three nontrivial 2-torsion points are

$$T_1 = (0, 0), \quad T_2 = (-1, 0), \quad T_3 = (-q^2, 0), \quad (3)$$

giving $(\mathbb{Z}/2\mathbb{Z})^2 \subseteq E_{\text{PCP}}(q)(\mathbb{Q})_{\text{tor}}$ for every $q \in \mathbb{Q}^* \setminus \{0, \pm 1\}$.

2.2. Rational four-torsion. Substituting $X = q$ into (1) yields $Y^2 = q(q + 1)(q + q^2) = q^2(q + 1)^2$, so $Y = \pm q(q + 1)$; substituting $X = -q$ yields $Y^2 = q^2(q - 1)^2$, so $Y = \pm q(q - 1)$. The duplication formula on (1) gives, with $a_2 = 1 + q^2$, $a_4 = q^2$ and $P = (q, q(q + 1))$, the slope $\lambda = q + 1$ and

$$X(2P) = \lambda^2 - a_2 - 2q = 0, \quad Y(2P) = \lambda(q - X(2P)) - q(q + 1) = 0, \quad (4)$$

an identity in $\mathbb{Q}(q)$, so $2P = (0, 0)$. The same computation with the sign variants gives $2 \cdot (\pm q, \pm q(q \pm 1)) = (0, 0)$. Hence the four points $(\pm q, \pm q(q \pm 1))$ have order 4 and

$$\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \subseteq E_{\text{PCP}}(q)(\mathbb{Q})_{\text{tor}} \quad \text{for all } q \in \mathbb{Q}^* \setminus \{0, \pm 1\}. \quad (5)$$

2.3. The classification (after Yoshida). The torsion is exactly $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and this is Yoshida's result transported to the present curve.

Proposition 1 (Torsion classification; cf. Yoshida [3, Lemma 2.1]). *For every Pythagorean $q \in \mathbb{Q}^* \setminus \{0, \pm 1\}$,*

$$E_{\text{PCP}}(q)(\mathbb{Q})_{\text{tor}} = \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

the eight points being the identity, the three of (3), and the four $(\pm q, \pm q(q \pm 1))$.

Proof (recalled). Containment is (5). For maximality, Mazur's theorem [2] leaves only $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ as supergroups of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$; the middle option fails Lagrange ($8 \nmid 12$), so it suffices to exclude a rational point of order 8. Such a point P has $2P$ of order 4, hence

$X(2P) \in \{q, -q\}$. Writing $X(2P) = q$ via the duplication formula and clearing denominators gives the quartic

$$G_1(X, q) = X^4 - 4qX^3 - (4q^3 + 2q^2 + 4q)X^2 - 4q^3X + q^4,$$

irreducible over $\mathbb{Q}[X, q]$. Since $G_1(0, q) = q^4 \neq 0$, any rational root is nonzero; setting $X = qZ$ and dividing by q^3 reduces $G_1 = 0$ to the quadratic in q

$$4Z^2q^2 - (Z^4 - 4Z^3 - 2Z^2 - 4Z + 1)q + 4Z^2 = 0,$$

whose discriminant is

$$\Delta(Z) = (Z^4 - 4Z^3 - 2Z^2 - 4Z + 1)^2 - 64Z^4 = (Z - 1)^4(Z + 1)^2(Z^2 - 6Z + 1). \quad (6)$$

The prefactor $(Z - 1)^4(Z + 1)^2 = ((Z - 1)^2(Z + 1))^2$ is a square, so $\Delta(Z)$ is a square exactly when $Z^2 - 6Z + 1 = w^2$. This conic has the rational point $(0, 1)$; parametrising by $w = tZ + 1$ gives $Z = Z(t) = -2(t + 3)/((t - 1)(t + 1))$, and back-substitution simplifies the recovered q to

$$q = q_+(t) = \left(\frac{4(t+1)}{(t-1)(t+3)} \right)^2 = s^2, \quad q_-(t) = 1/s^2.$$

Thus any order-eight parameter forces $q = s^2$ or $1/s^2$. The Pythagorean hypothesis $1 + q^2 \in \mathbb{Q}^2$ then demands $s^4 + 1 \in \mathbb{Q}^2$, i.e. a rational point on $u^2 = s^4 + 1$. This curve is birational to $y^2 = x^3 + 4x$ (conductor 32, Cremona **32a1**), of analytic rank 0 and torsion $\mathbb{Z}/4\mathbb{Z}$; by Kolyvagin its Mordell–Weil rank is 0, so the only rational solution is $s = 0$, giving $q \in \{0, \infty\}$, both excluded. The case $X(2P) = -q$ is identical with $q \mapsto -q$. The reduction to $s^4 + 1 = u^2$ and its resolution by Fermat’s theorem on right triangles is exactly the descent of Yoshida [3, Lemma 2.1], there phrased on $y^2 = x^3 - x$; the discriminant identity (6) is the only presentational change. $\square$

Remark 1. Equation (6) is the single point at which the present write-up departs from [3] at the level of formulae: it exhibits the order-eight obstruction as a uniform discriminant in the parameter q , with the square prefactor stripped off, rather than passing through an auxiliary curve. It is a convenience, not a new theorem.

3. DEGENERACY OF TORSION UNDER THE RECOVERY MAP

The recovery map (2) is regular on the affine locus $X \neq \pm q$ and has pole divisor $\{X = q\} \cup \{X = -q\}$. We record its action on torsion. This is the new content of the note.

Theorem 1 (Torsion degeneracy). *Let $q \in \mathbb{Q}^* \setminus \{0, \pm 1\}$ be Pythagorean. Then every one of the seven non-identity rational torsion points of $E_{\text{PCP}}(q)$ is sent by φ into $\{0, \infty\}$: the three 2-torsion points (3) map to $\varphi = 0$, and the four order-four points $(\pm q, \pm q(q \pm 1))$ lie on the pole divisor of φ , hence map to ∞ . (The identity O is the point at infinity of $E_{\text{PCP}}(q)$, outside the affine domain of φ .) Consequently no rational torsion point of $E_{\text{PCP}}(q)$ has finite nonzero image under φ .*

Proof. By Proposition 1 the torsion consists of the identity, the three points of (3), and the four order-four points.

For a 2-torsion point $T_i = (X_i, 0)$ the numerator $2Yq$ of (2) vanishes, while the denominator $q^2 - X_i^2$ does not: for T_1 it is $q^2 \neq 0$, for T_2 it is $q^2 - 1 \neq 0$ (as $q \neq \pm 1$), and for T_3 it is $q^2 - q^4 = q^2(1 - q^2) \neq 0$. Hence $\varphi(T_i) = 0$ for $i = 1, 2, 3$.

For an order-four point $(\pm q, \pm q(q \pm 1))$ the X -coordinate satisfies $X^2 = q^2$, so the denominator $q^2 - X^2$ is 0, while the numerator $2Yq = \pm 2q^2(q \pm 1)$ is nonzero because $q \notin \{0, \pm 1\}$. Each of the four points therefore lies on the pole divisor, and φ takes the value ∞ there.

The identity O is the point at infinity of $E_{\text{PCP}}(q)$, outside the affine domain of φ , and so is not assigned a finite nonzero edge; it is excluded from the seven counted above. The two displayed evaluations are identities in $\mathbb{Q}(q)$, independent of the particular Pythagorean value of q . $\square$

Corollary 1. *On any Pythagorean fibre, a rational point of $E_{\text{PCP}}(q)$ decoding to a finite nonzero cuboid edge is necessarily of infinite order. In particular the torsion subgroup may be discarded at the outset of any search for perfect-cuboid solutions on the fibre.*

Proof. A finite nonzero value of φ excludes, by Theorem 1, every torsion point. Since $E_{\text{PCP}}(q)(\mathbb{Q}) = E_{\text{PCP}}(q)(\mathbb{Q})_{\text{tor}} \oplus \mathbb{Z}^r$ with the torsion exhausted by Proposition 1, such a point has infinite order. $\square$

Remark 2. Yoshida [3] has no analogue of Theorem 1: his map runs from non-torsion points to face cuboids, producing solutions, whereas φ here detects the *absence* of solutions among torsion. The two viewpoints are compatible – a perfect cuboid is a face cuboid with one further squareness condition – and the present statement is the structural reason the torsion of (1) contributes nothing to the perfect-cuboid count.

4. COMPUTATIONAL VERIFICATION

All claims were checked symbolically in `sympy` and numerically in `PARI/GP`; the scripts and their output accompany this note.

- (1) *Discriminant identity* (6). The reduction $X(2P) = q \Rightarrow G_1 = 0$, the substitution $X = qZ$, and the factorisation $\Delta(Z) = (Z-1)^4(Z+1)^2(Z^2-6Z+1)$ were verified as exact identities, together with the conic parametrisation giving $q = s^2$ (`03_discriminant_identity.py`).
- (2) *Degeneracy* (Theorem 1). The action of φ on the eight torsion points was confirmed as an identity in $\mathbb{Q}(q)$ (`05_recovery_degeneracy.py`) and numerically over 62 Pythagorean values of q , all $62 \times 8 = 496$ evaluations landing in $\{0, \infty\}$ with torsion $[4, 2]$ in every case (`02_torsion_sweep.gp`).
- (3) *Fermat obstruction*. The curve $y^2 = x^3 + 4x$ has analytic rank 0 and torsion $\mathbb{Z}/4\mathbb{Z}$, so $u^2 = s^4 + 1$ has only $s = 0$ (`04_fermat_curve.gp`).
- (4) *Twist, not isomorphism*. For Pythagorean $q \in \{4/3, 3/4, 5/12, 12/5, 15/8, 8/15\}$ no rational s of height ≤ 80 has $j(E_{1,s}) = j(E_{\text{PCP}}(q))$, whereas the j -invariants agree along the non-Pythagorean locus $q = (s^2 + 1)/(2s)$; the curves are -1 -twist-related (`01_isomorphism.gp`).

5. SCOPE

This is a short structural note. Its single new theorem, the torsion degeneracy 1, is a companion observation to Yoshida [3]: it identifies the recovery map under which the entire torsion subgroup of the perfect-cuboid family is degenerate, and thereby justifies restricting any subsequent rank-, descent-, or Chabauty-type search to the non-torsion locus (Corollary 1). The torsion classification itself is Yoshida’s and is presented as such. We make no claim about the existence or non-existence of perfect cuboids; the content here is the elementary but, to our knowledge, previously unrecorded fact that torsion alone never produces a finite nonzero edge.

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